THIRD YEAR CSE PROJECT | MACHINE LEARNING & CYBERSECURITY

Intelligent Network Traffic Monitoring System

DDoS Attack Detection using Ensemble Machine Learning | CIC-DDoS2019 Dataset

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PROBLEM STATEMENT

Distributed Denial-of-Service (DDoS) attacks flood network servers with malicious traffic, rendering services unavailable. Traditional rule-based firewalls fail against modern, high-volume, multi-vector attacks.

- Over 17 million DDoS attacks occur globally every year
- Attack traffic mimics legitimate patterns — hard to detect
- Manual analysis cannot scale to Gbps network speeds
- ML can learn statistical traffic patterns automatically

DATASET: CIC-DDoS2019

Canadian Institute for Cybersecurity benchmark dataset captured on a realistic network testbed. Contains labelled benign and 11 DDoS attack types.

Source Files:	17 Parquet files
Raw Records:	500,000+
After Cleaning:	426,076 rows
Raw Features:	88 columns
Selected Feat.:	30 best features
Train Set:	340,860 rows (80%)
Post-SMOTE:	530,194 rows (balanced)
Test Set:	85,216 rows (20%, held-out)
Attack Types:	SYN, UDP, DNS, LDAP, MSSQL, NTP...

PREPROCESSING PIPELINE

01 Load 17 Parquet files with pyarrow
02 Drop null / infinite / duplicate rows
03 Binary labels: BENIGN=0, ATTACK=1
04 Feature selection: 88 -> 30 features (RandomForest importance + correlation)
05 80/20 stratified train-test split
06 StandardScaler on X_train only
07 SMOTE on training set only (balance classes 50%-50%)

30 SELECTED FEATURES (TOP 12)

- Flow Duration
- Fwd Packet Length Max
- Fwd Packet Length Mean
- Bwd Packet Length Max
- Fwd Packets/s
- Bwd Packets/s
- Flow IAT Mean
- Flow IAT Max
- Fwd Header Length
- Fwd IAT Total
- Packet Length Mean
- Protocol

5 MODELS TRAINED

Decision Tree — Baseline	Acc: 0.9992
Random Forest — Bagging (100 trees)	Acc: 0.9993
XGBoost — Boosting (100 rounds)	Acc: 0.9990
Voting Ensemble — Soft voting DT+RF+XGB	Acc: 0.9993
Stacking Ensemble — Meta-Learning (LR)	Acc: 0.9994

BEST MODEL: Stacking Ensemble | Accuracy 99.94% | F1: 0.9996

PERFORMANCE METRICS (TEST SET: 85,216 SAMPL

Model	Accuracy	Precision	Recall	F
Decision Tree	99.92%	99.97%	99.93%	99.
Random Forest	99.93%	99.99%	99.93%	99.
XGBoost	99.90%	99.99%	99.88%	99.94%
Voting Ensemble	99.93%	99.99%	99.92%	99.
Stacking Ens.	**99.94%**	**99.99%**	**99.93%**	**99.**

CONFUSION MATRIX (STACKING ENSEMBLE — BE

	Predicted BENIGN	Predicted ATTACK
	18,932 True Negative (TN)	9 False Positive (FP)
	46 False Negative (FN)	66,229 True Positive (TP)

Test size: 8
Benign: 18,941
Attack: 66,

FP=9 (false
FN=46 (mis

WHY ENSEMBLE LEARNING?

- Single models are prone to overfitting specific patterns
- Voting combines DT + RF + XGBoost — reduces individual bias
- Stacking uses meta-learner (Logistic Regression) to optimally weight base model predictions
- Ensemble achieves lowest FN (46) — fewest missed attacks
- Bagging (RF): reduces variance by averaging 100 trees
- Boosting (XGBoost): corrects errors of previous trees sequentially
- Meta-Learning: learns WHEN to trust each base model

KEY RESULTS AT A GLANCE

426,076	30	5	99.94%
Total Records	Features Used	Models Trained	Best Accuracy

FLASK WEB DASHBOARD

URL: http://127.0.0.1:5000

- **Dashboard** — Stat cards, 10-step pipeline flow, best model summary
- **Traffic Prediction** — CSV upload -> real ML inference -> BENIGN/ATTACK table
- **Model Comparison** — Side-by-side metrics, bar charts, radar chart
- **Evaluation** — Confusion matrices for all 5 models
- **About** — Dataset, pipeline, tech stack, limitations

Real ML Inference | No hardcoded results | CSV Download support

TECHNOLOGY STACK

Python 3.11	Core language
scikit-learn 1.9	DT, RF, Stacking, StandardScaler
XGBoost 3.4	Gradient boosting
pandas / NumPy	Data processing
imbalanced-learn	SMOTE oversampling
Flask 3.1.3	Web framework
Chart.js 4.4	Interactive charts
joblib 1.6	Model serialization (.pkl)

RESULTS & IMPACT

- Stacking Ensemble achieves 99.94% accuracy on 85,216 test samples
- Only 9 false positives — reduces unnecessary blocking of legitimate users
- Only 46 missed attacks (FN) — critical for security reliability
- SMOTE resolved 78%-22% class imbalance without touching test data
- 30 features selected from 88 — improves speed with no accuracy loss
- CSV-based demo: teacher can upload any 30-feature

LIMITATIONS & FUTURE SCOPE

Current Limitations:

- Offline only — no live packet capture
- Binary classification (not multi-class)
- Feature selection performed pre-split (mild leakage)
- Tested only on CIC-DDoS2019 dataset

Future Scope:

- Integrate with scapy for live packet capture
- Extend to 11-class multi-label DDoS classification
- Deploy on cloud (AWS/GCP) with Docker + gunicorn
- Add auto-retraining pipeline for data drift detection